

2 — Simulation: families, prior, constraints

1 Generation

For each family we sweep over $(\bar{n}, \tilde{\delta})$ (expected count and departure from CSR); train and test sets are carved from this pool of clouds.

Parameterisation

- Each θ splits into $\bar{n}$, a *shape* (the length scale the structure lives at) and an *amplitude* (how strong it is at that scale).
- At fixed $\bar{n}$ and shape the amplitude moves $\tilde{\delta}$: it is monotone in the amplitude (checked on grids).
- Amplitude $\rightarrow 0$ recovers CSR.

Shape/amplitude parameters for each family:

- **Poisson**: homogeneous, intensity $\bar{n}$. No structure ($\tilde{\delta} = 0$)
- **Thomas**: parents at intensity κ , each with $\text{Poisson}(\mu)$ offspring displaced by $N(0, \sigma^2 I)$. Shape σ , amplitude μ (with $\kappa = \bar{n}/\mu$, so $\mu \rightarrow 0$ is a superposition of many one-point parents, i.e. $\text{Poisson}(\bar{n})$)
- **Nested**: same construction at two levels: meta-parents at κ , $\text{Poisson}(\mu_1)$ parents at scale σ_1 , $\text{Poisson}(\mu_2)$ children at scale σ_2 . Shape $(\sigma_2, \rho = \sigma_1/\sigma_2, \mu_1)$, amplitude μ_2
- **LGCP**: $\Lambda = e^{\mu+Y}$ with $\text{Cov } Y = \sigma^2 e^{-r/s}$. Shape s , amplitude σ^2
- **Matérn II**: primaries at λ_p , thinned by a hard core of radius R . No shape; amplitude R

Draw order:

- $\bar{n} \sim \text{log-uniform}[100, 800]$
- $\tilde{\delta} \sim \text{log-uniform}[0.1, 4]$ (Poisson: skip the rest)
- Sweep shape on log-uniform intervals allowed by constraints

Constraints

Four rules drop parameters for which $\tilde{\delta}$ would be a misleading label. All use the window $[r_{\min}(\bar{n}), 0.25]$, with $r_{\min}(\bar{n})$ set at 10 expected pairs.

- **Structure sits inside the window.** Structure outside the window is invisible to $\tilde{\delta}$. So $\sigma \leq 0.098$ (Thomas), $s \leq 0.155$ (LGCP), and matching bounds on the two nested scales
- **Nested levels stay distinguishable.** Otherwise nested is just Thomas: scales separated ($\rho \geq 2.49$) and both clustering non-negligibly ($1 \leq \mu_1 \leq 4$)
- **Counts stay near $\bar{n}$.** $\tilde{\delta}$ is defined at $\bar{n}$, so a few large clusters swinging n breaks the label: CV of n (closed form) at most 0.45
- **Matérn packing.** Type II thinning cannot pass a fill of $\pi R^2 \bar{n} = 1$ and degenerates near it; capped at 0.95

Range of $\tilde{\delta}$

- The constraints cap how much departure each family can reach. Largest attainable $\tilde{\delta}$:

$\bar{n}$	100	152	230	348	528	800
Thomas	16.4	26.4	41.6	64.6	99.9	154
Nested	14.1	23.4	38.0	60.9	96.0	149
LGCP	16.9	26.9	41.6	63.1	94.5	147
Matérn II	4.2	5.2	6.4	7.9	9.8	11.8

- Common range at all $\bar{n}$ is $\tilde{\delta} \in [0.1, 4]$, inside which $\bar{n}$ and $\tilde{\delta}$ can be drawn independently.

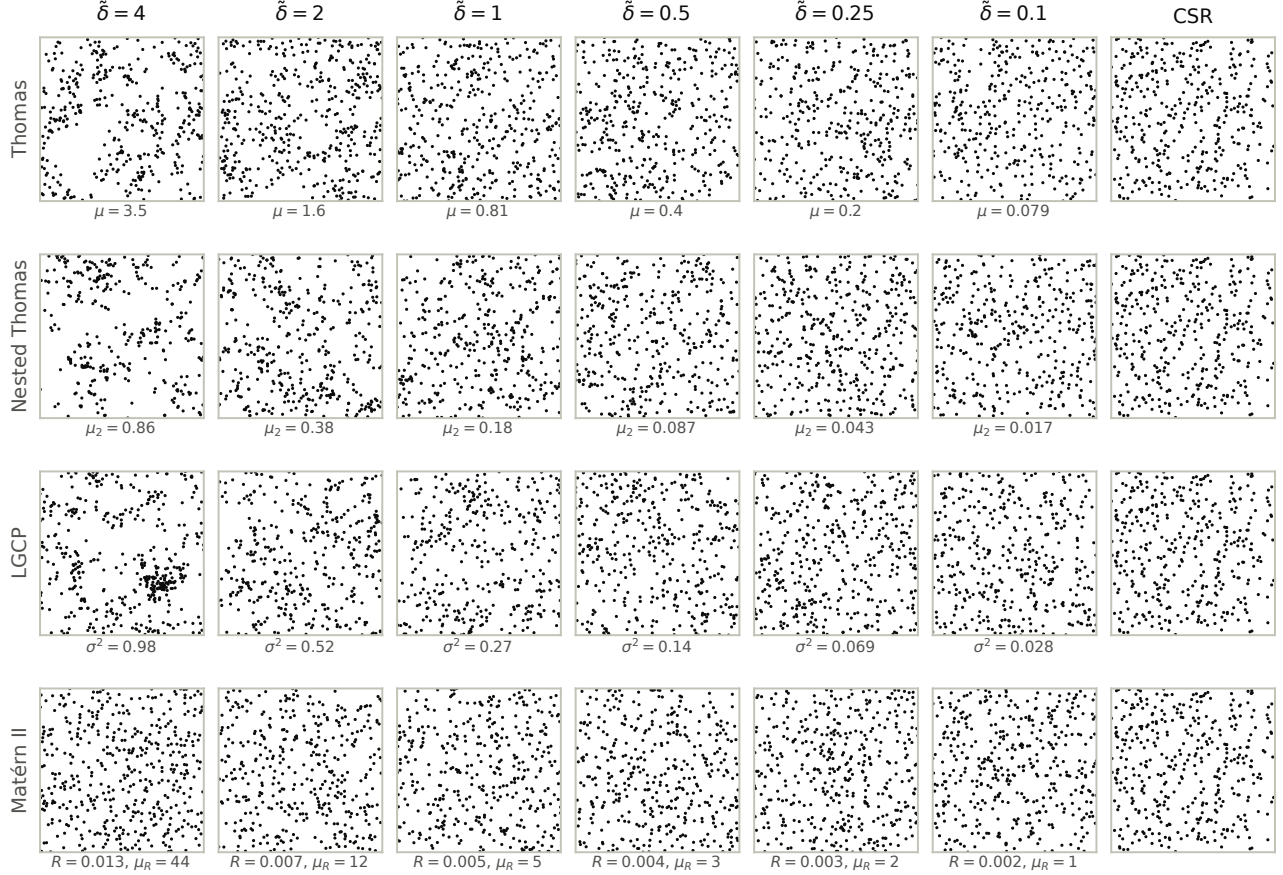


Figure 1: One pattern per cell, $\bar{n} = 400$, fixed shape per family (Thomas $\sigma = 0.03$; nested $\sigma_2 = 0.008$, $\rho = 5$, $\mu_1 = 3$; LGCP $s = 0.04$). Amplitude changes along rows (value stated under each panel). For Matérn that is the hard core R , with μ_R , the CSR pairs it forbids: the row looks like CSR throughout because even at $\tilde{\delta} = 4$ the hard core is about a quarter of the mean spacing $1/\sqrt{\bar{n}} = 0.05$.